ADVANCED COMPUTER NETWORKS-PCS223

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ME - CSE1

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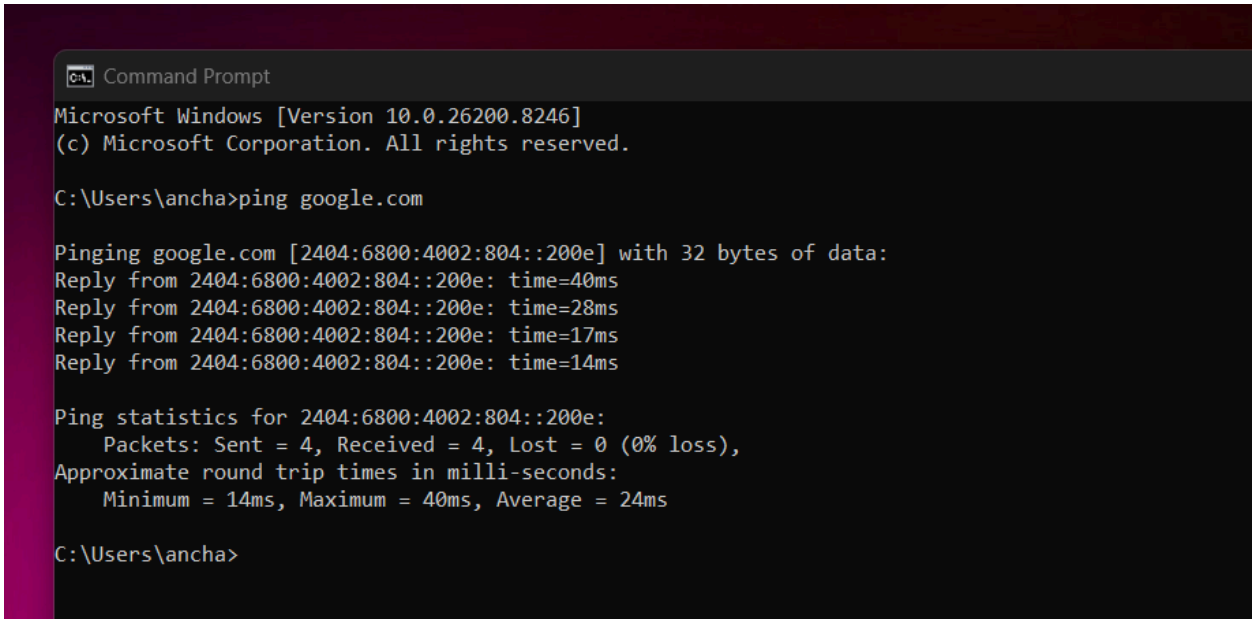
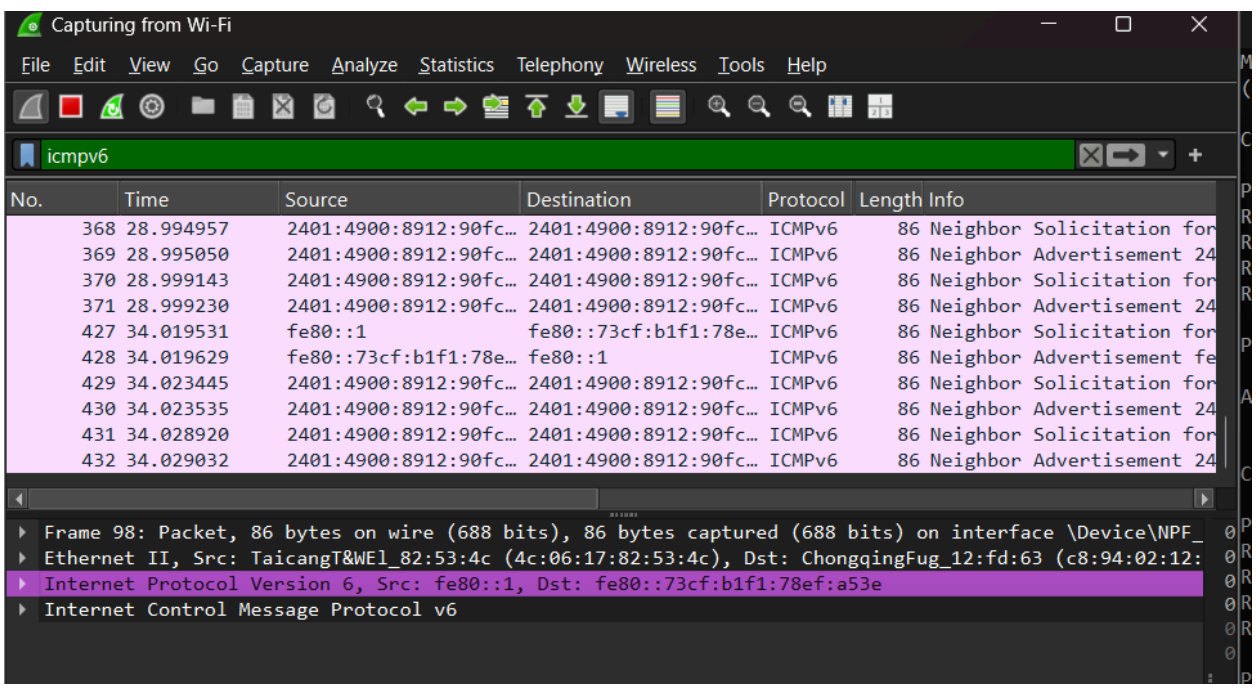
Assignment 2

Cmd-



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ancha>ping google.com

Pinging google.com [2404:6800:4002:804::200e] with 32 bytes of data:
Reply from 2404:6800:4002:804::200e: time=40ms
Reply from 2404:6800:4002:804::200e: time=28ms
Reply from 2404:6800:4002:804::200e: time=17ms
Reply from 2404:6800:4002:804::200e: time=14ms

Ping statistics for 2404:6800:4002:804::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 40ms, Average = 24ms

C:\Users\ancha>
```

Wireshark - Via Wifi



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

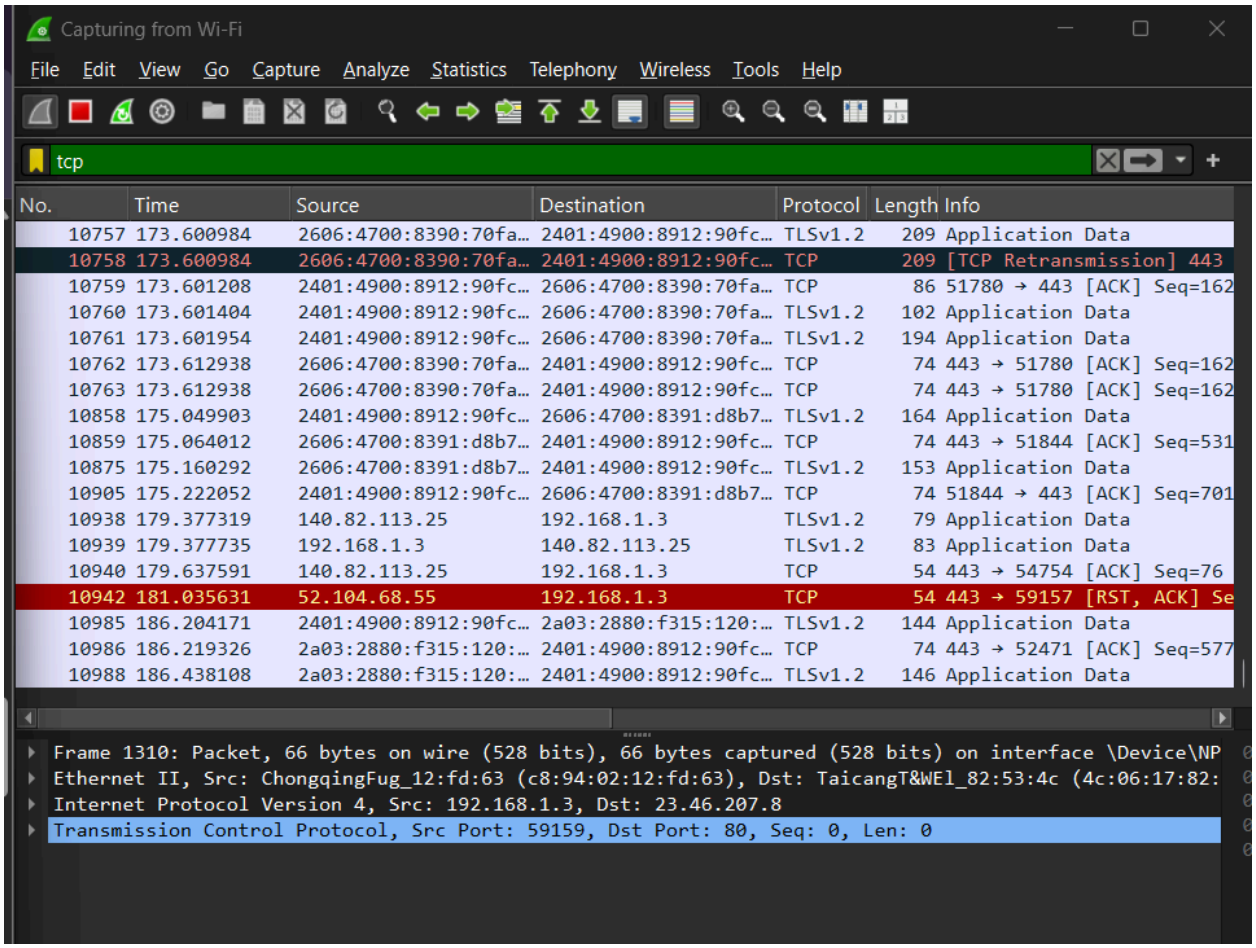
icmpv6

No.	Time	Source	Destination	Protocol	Length	Info
368	28.994957	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
369	28.995050	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24
370	28.999143	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
371	28.999230	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24
427	34.019531	fe80::1	fe80::73cf:b1f1:78e…	ICMPv6	86	Neighbor Solicitation for
428	34.019629	fe80::73cf:b1f1:78e…	fe80::1	ICMPv6	86	Neighbor Advertisement fe
429	34.023445	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
430	34.023535	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24
431	34.028920	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
432	34.029032	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24

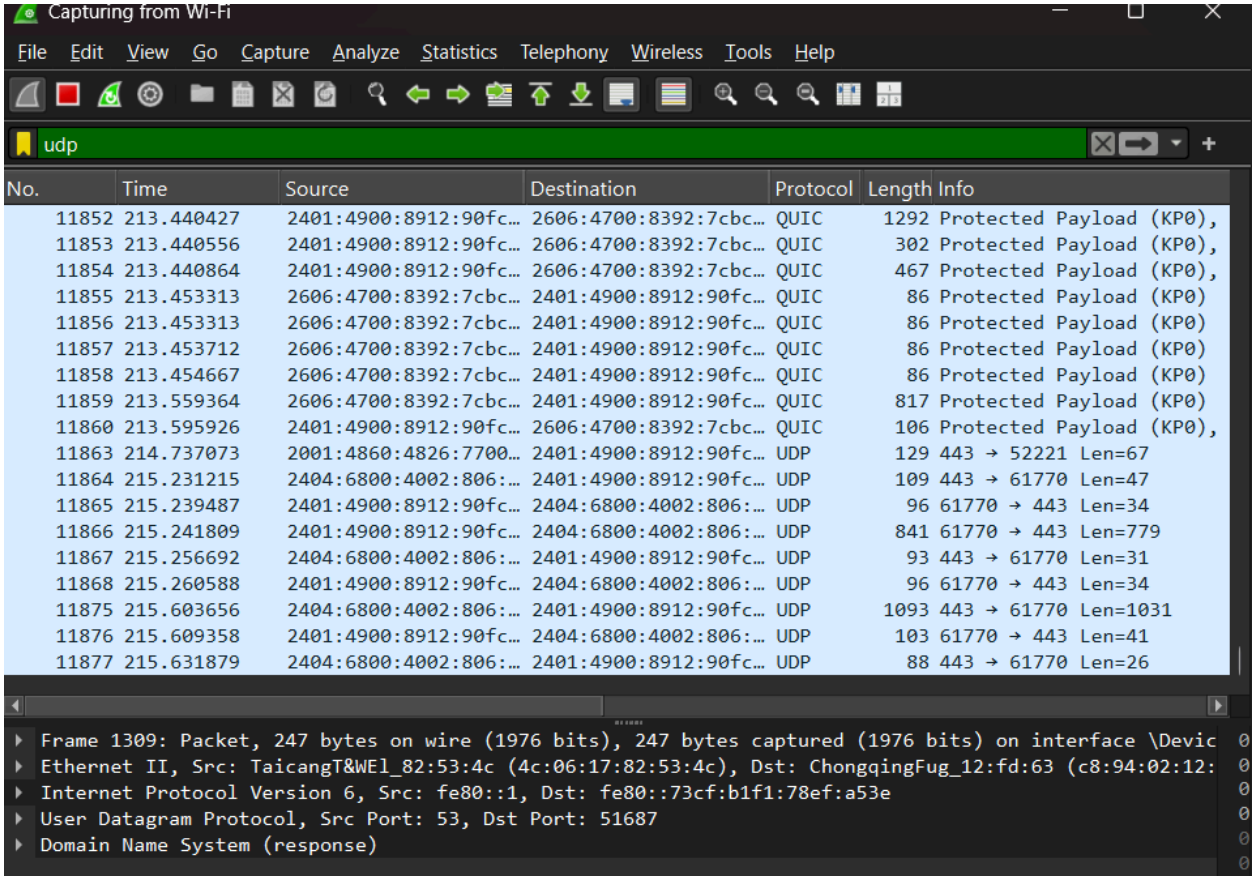
```
Frame 98: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:82:53:4c), Dst: ChongqingFug_12:fd:63 (c8:94:02:12:
Internet Protocol Version 6, Src: fe80::1, Dst: fe80::73cf:b1f1:78ef:a53e
Internet Control Message Protocol v6
```

2.

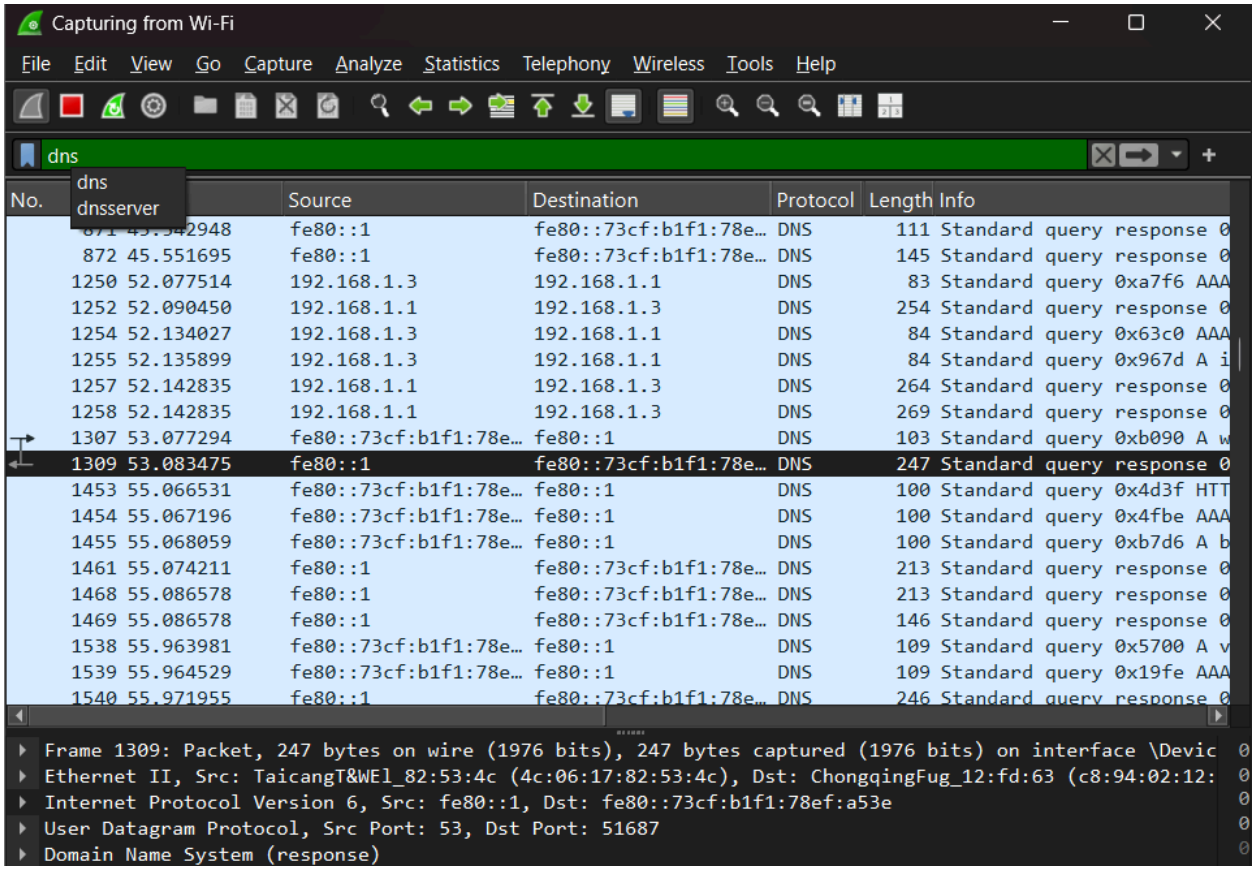
Tcp


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
10757 173.600984 2606:4700:8390:70fa… 2401:4900:8912:90fc… TLSv1.2 209 Application Data
10758 173.600984 2606:4700:8390:70fa… 2401:4900:8912:90fc… TCP 209 [TCP Retransmission] 443
10759 173.601208 2401:4900:8912:90fc… 2606:4700:8390:70fa… TCP 86 51780 → 443 [ACK] Seq=162
10760 173.601404 2401:4900:8912:90fc… 2606:4700:8390:70fa… TLSv1.2 102 Application Data
10761 173.601954 2401:4900:8912:90fc… 2606:4700:8390:70fa… TLSv1.2 194 Application Data
10762 173.612938 2606:4700:8390:70fa… 2401:4900:8912:90fc… TCP 74 443 → 51780 [ACK] Seq=162
10763 173.612938 2606:4700:8390:70fa… 2401:4900:8912:90fc… TCP 74 443 → 51780 [ACK] Seq=162
10858 175.049903 2401:4900:8912:90fc… 2606:4700:8391:d8b7… TLSv1.2 164 Application Data
10859 175.064012 2606:4700:8391:d8b7… 2401:4900:8912:90fc… TCP 74 443 → 51844 [ACK] Seq=531
10875 175.160292 2606:4700:8391:d8b7… 2401:4900:8912:90fc… TLSv1.2 153 Application Data
10905 175.222052 2401:4900:8912:90fc… 2606:4700:8391:d8b7… TCP 74 51844 → 443 [ACK] Seq=701
10938 179.377319 140.82.113.25 192.168.1.3 TLSv1.2 79 Application Data
10939 179.377735 192.168.1.3 140.82.113.25 TLSv1.2 83 Application Data
10940 179.637591 140.82.113.25 192.168.1.3 TCP 54 443 → 54754 [ACK] Seq=76
10942 181.035631 52.104.68.55 192.168.1.3 TCP 54 443 → 59157 [RST, ACK] Se
10985 186.204171 2401:4900:8912:90fc… 2a03:2880:f315:120:… TLSv1.2 144 Application Data
10986 186.219326 2a03:2880:f315:120:… 2401:4900:8912:90fc… TCP 74 443 → 52471 [ACK] Seq=577
10988 186.438108 2a03:2880:f315:120:… 2401:4900:8912:90fc… TLSv1.2 146 Application Data
Frame 1310: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NP
Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:12:fd:63), Dst: TaicangT&WEl_82:53:4c (4c:06:17:82:
Internet Protocol Version 4, Src: 192.168.1.3, Dst: 23.46.207.8
Transmission Control Protocol, Src Port: 59159, Dst Port: 80, Seq: 0, Len: 0

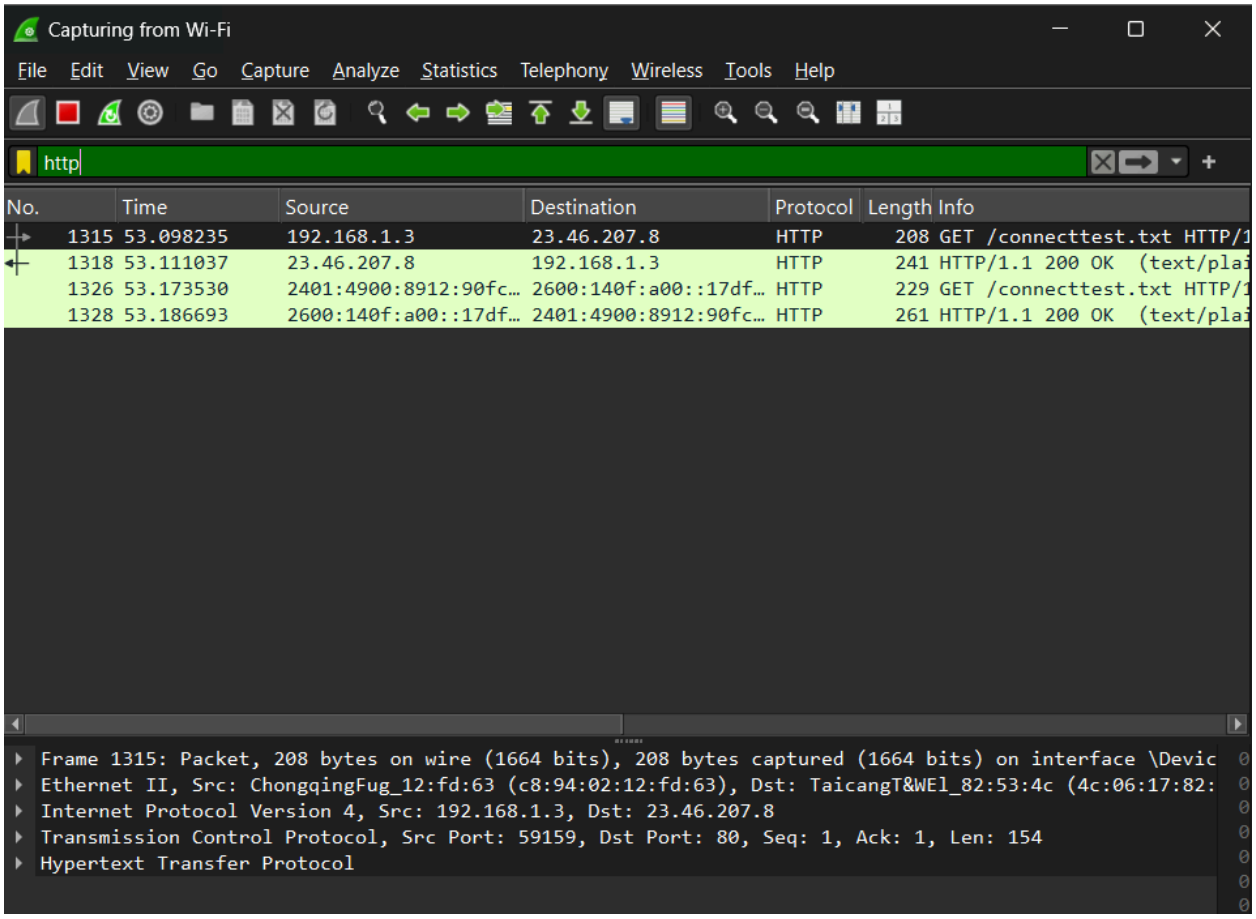
3.UDP


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
11852 213.440427 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 1292 Protected Payload (KP0),
11853 213.440556 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 302 Protected Payload (KP0),
11854 213.440864 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 467 Protected Payload (KP0),
11855 213.453313 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11856 213.453313 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11857 213.453712 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11858 213.454667 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11859 213.559364 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 817 Protected Payload (KP0)
11860 213.595926 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 106 Protected Payload (KP0),
11863 214.737073 2001:4860:4826:7700… 2401:4900:8912:90fc… UDP 129 443 → 52221 Len=67
11864 215.231215 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 109 443 → 61770 Len=47
11865 215.239487 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 96 61770 → 443 Len=34
11866 215.241809 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 841 61770 → 443 Len=779
11867 215.256692 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 93 443 → 61770 Len=31
11868 215.260588 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 96 61770 → 443 Len=34
11875 215.603656 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 1093 443 → 61770 Len=1031
11876 215.609358 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 103 61770 → 443 Len=41
11877 215.631879 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 88 443 → 61770 Len=26
Frame 1309: Packet, 247 bytes on wire (1976 bits), 247 bytes captured (1976 bits) on interface \Devic
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:82:53:4c), Dst: ChongqingFug_12:fd:63 (c8:94:02:12:
Internet Protocol Version 6, Src: fe80::1, Dst: fe80::73cf:b1f1:78ef:a53e
User Datagram Protocol, Src Port: 53, Dst Port: 51687
Domain Name System (response)

Dns -


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
dns
dnsserver
No. Source Destination Protocol Length Info
872 45.551695 fe80::1 fe80::73cf:b1f1:78e… DNS 145 Standard query response 0
1250 52.077514 192.168.1.3 192.168.1.1 DNS 83 Standard query 0xa7f6 AAA
1252 52.090450 192.168.1.1 192.168.1.3 DNS 254 Standard query response 0
1254 52.134027 192.168.1.3 192.168.1.1 DNS 84 Standard query 0x63c0 AAA
1255 52.135899 192.168.1.3 192.168.1.1 DNS 84 Standard query 0x967d A i
1257 52.142835 192.168.1.1 192.168.1.3 DNS 264 Standard query response 0
1258 52.142835 192.168.1.1 192.168.1.3 DNS 269 Standard query response 0
1307 53.077294 fe80::73cf:b1f1:78e… fe80::1 DNS 103 Standard query 0xb090 A w
1309 53.083475 fe80::1 fe80::73cf:b1f1:78e… DNS 247 Standard query response 0
1453 55.066531 fe80::73cf:b1f1:78e… fe80::1 DNS 100 Standard query 0x4d3f HTT
1454 55.067196 fe80::73cf:b1f1:78e… fe80::1 DNS 100 Standard query 0x4fbe AAA
1455 55.068059 fe80::73cf:b1f1:78e… fe80::1 DNS 100 Standard query 0xb7d6 A b
1461 55.074211 fe80::1 fe80::73cf:b1f1:78e… DNS 213 Standard query response 0
1468 55.086578 fe80::1 fe80::73cf:b1f1:78e… DNS 213 Standard query response 0
1469 55.086578 fe80::1 fe80::73cf:b1f1:78e… DNS 146 Standard query response 0
1538 55.963981 fe80::73cf:b1f1:78e… fe80::1 DNS 109 Standard query 0x5700 A v
1539 55.964529 fe80::73cf:b1f1:78e… fe80::1 DNS 109 Standard query 0x19fe AAA
Frame 1309: Packet, 247 bytes on wire (1976 bits), 247 bytes captured (1976 bits) on interface \Devic
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:82:53:4c), Dst: ChongqingFug_12:fd:63 (c8:94:02:12:
Internet Protocol Version 6, Src: fe80::1, Dst: fe80::73cf:b1f1:78ef:a53e
User Datagram Protocol, Src Port: 53, Dst Port: 51687
Domain Name System (response)

Http


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
1315 53.098235 192.168.1.3 23.46.207.8 HTTP 208 GET /connecttest.txt HTTP/1
1318 53.111037 23.46.207.8 192.168.1.3 HTTP 241 HTTP/1.1 200 OK (text/plai
1326 53.173530 2401:4900:8912:90fc… 2600:140f:a00::17df… HTTP 229 GET /connecttest.txt HTTP/1
1328 53.186693 2600:140f:a00::17df… 2401:4900:8912:90fc… HTTP 261 HTTP/1.1 200 OK (text/plai
Frame 1315: Packet, 208 bytes on wire (1664 bits), 208 bytes captured (1664 bits) on interface \Devic
Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:12:fd:63), Dst: TaicangT&WEl_82:53:4c (4c:06:17:82:
Internet Protocol Version 4, Src: 192.168.1.3, Dst: 23.46.207.8
Transmission Control Protocol, Src Port: 59159, Dst Port: 80, Seq: 1, Ack: 1, Len: 154
Hypertext Transfer Protocol

ARp

arp

No.	Time	Source	Destination	Protocol	Length	Info
14364	325.046482	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14365	325.046501	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14449	330.162927	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14450	330.162956	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14537	335.362047	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14538	335.362071	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14599	340.256540	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14600	340.256594	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14697	345.678398	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14698	345.678428	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14712	350.806909	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14713	350.806943	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14757	354.935177	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14758	354.935250	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14874	360.099860	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14875	360.099880	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14974	365.212646	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14975	365.212676	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:

```
▸ Frame 1260: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NP
▸ Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:12:fd:63), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Address Resolution Protocol (ARP Probe)
```